

Leading University
Department of Computer Science and Engineering
CSE – 1151 (Discrete Mathematics)
Tutorial Exam: 01

Full Marks: 10 (5 x 2)

[Answer any 5 questions] Time: 20 mins

1. What is a proposition? Give examples of propositions and non-propositions.
2. Draw the circuit diagram: $(p \vee q) \wedge \neg r$.
3. What is the logical equivalent of the implication $p \leftrightarrow q$?
4. Write down the Double negation law of Logical Expression.
5. Calculate the bitwise OR of the binary numbers 1001010 and 0101011.
6. Which one has the highest and which one has the lowest precedence among $\wedge, \vee, \neg, \rightarrow, \leftrightarrow$?

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1. What is a proposition? Is the statement “ $2 + 2 = 3$ ” a proposition? (Yes/No).
2. Draw the circuit diagram: $(p \wedge \neg q) \vee \neg r$.
3. What is the logical equivalent of the implication $p \rightarrow q$?
4. Write down De Morgan's laws of Logical Expression.
5. Calculate the bitwise XOR of the binary numbers 1001010 and 0101011.
6. What is a contradiction?